Problem Set 3

Applied Stats/Quant Methods 1
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Due: November 13, 2025

Instructions

- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

* Note: "Spending" variable, `difflog`, is the difference in log of spending between incumbent and challenger.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

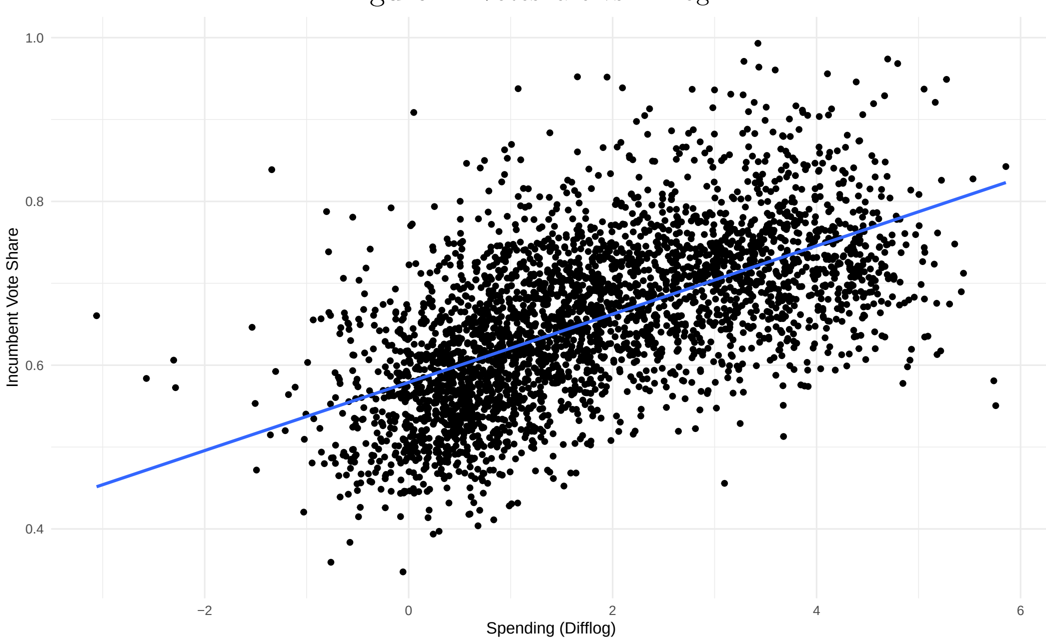
```
reg_line_1 <- lm(inc_sub$voteshare ~ inc_sub$difflog)
```

Table 1: Incumbent Vote Share Explained by Spending

	Dependent variable:
	Voteshare Coefficients (Reg 1)
Spending (Difflog)	0.042***
	(0.001)
Constant	0.579***
	(0.002)
Observations	3,193
R^2	0.367
Adjusted R^2	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 1: Voteshare vs Difflog.



3. Save the residuals of the model in a separate object.

```r
residuals_rl_1 <- reg_line_1$residuals
```

4. Write the prediction equation.

$$\hat{Y} = \alpha_1 + \beta_1 X = 0.579 + 0.042X$$

$$\widehat{\texttt{voteshare}} = 0.579 + 0.042\texttt{difflog}$$

In R:

```r
alpha_rl_1 <- reg_line_1$coefficients[1]
beta_rl_1 <- reg_line_1$coefficients[2]

# y = alpha_rl_1 + beta_rl_1*x

p_vs_fun <- function(x) {
  y <- alpha_rl_1 + beta_rl_1*x
  return(y)
}
```

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

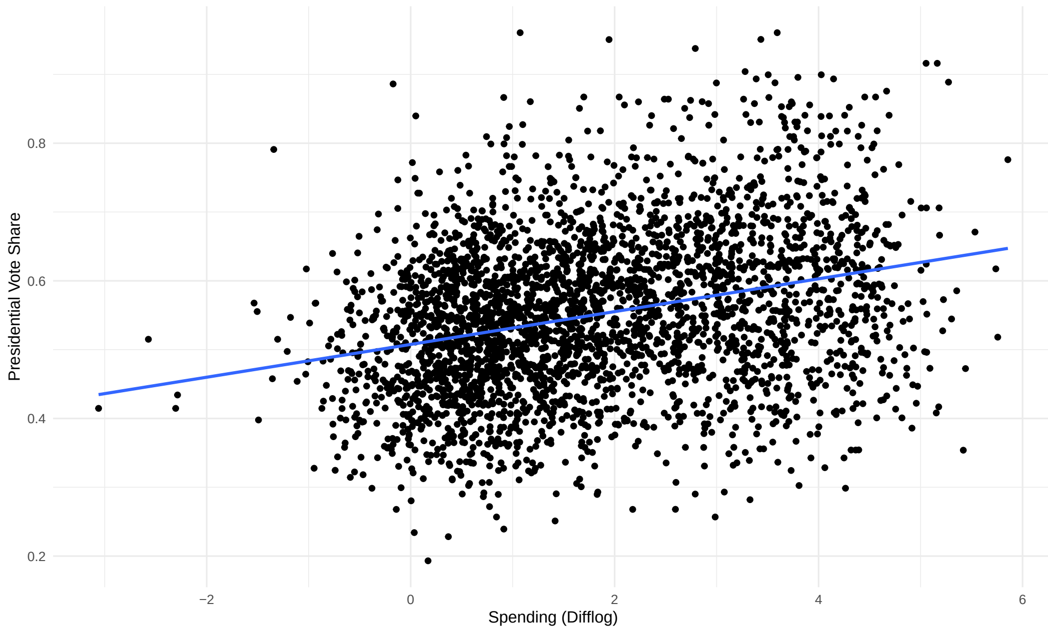
```
reg_line_2 <- lm(inc_sub$presvote ~ inc_sub$difflog)
```

Table 2: Presidental Vote Share Explained by Spending

	Dependent variable:
	Presvote Coefficients (Reg 2)
Spending (Difflog)	0.024***
	(0.001)
Constant	0.508***
	(0.003)
Observations	3,193
R^2	0.088
Adjusted R^2	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 2: Presvote vs Difflog.



3. Save the residuals of the model in a separate object.

```
residuals_rl_2 <- reg_line_2$residuals
```

4. Write the prediction equation.

$$\hat{Y} = \alpha_2 + \beta_2 X = 0.508 + 0.024X$$

$$\widehat{\texttt{presvote}} = 0.508 + 0.024\texttt{difflog}$$

In R:

```
alpha_rl_2 <- reg_line_2$coefficients[1]
beta_rl_2 <- reg_line_2$coefficients[2]

# y = alpha_rl_2 + beta_rl_2*x

p_vs_fun_2 <- function(x) {
  y <- alpha_rl_2 + beta_rl_2*x
  return(y)
}
```

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

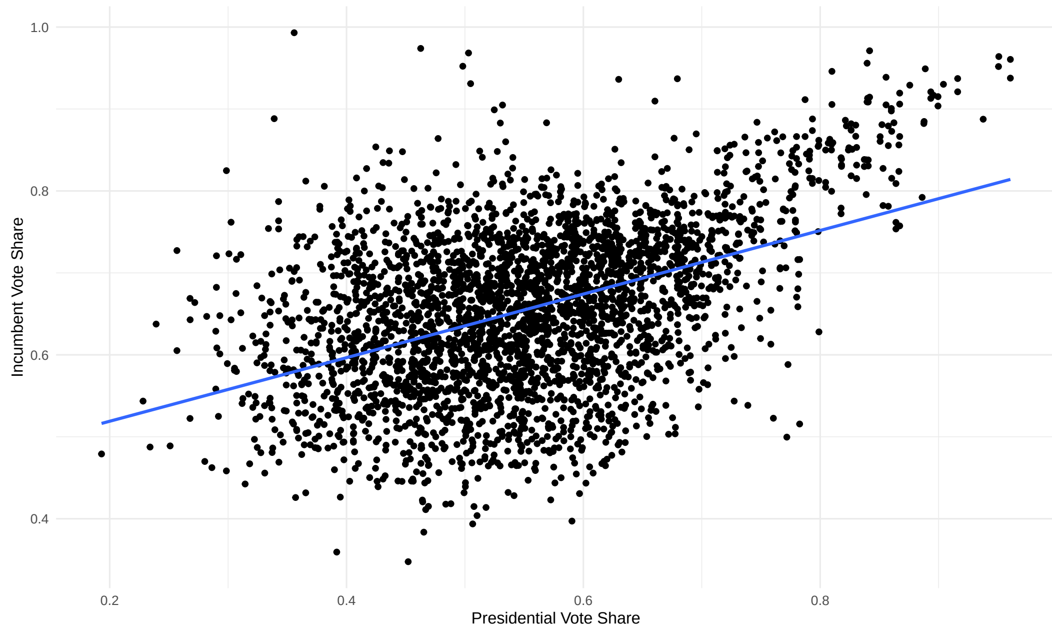
```
reg_line_3 <- lm(inc_sub$voteshare ~ inc_sub$presvote)
```

Table 3: Incumbent Vote Share Explained by Presidential Vote Share

	Dependent variable:
	Voteshare Coefficients (Reg 3)
Presvote	0.388***
	(0.013)
Constant	0.441***
	(0.008)
Observations	3,193
R^2	0.206
Adjusted R^2	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two variables and add the regression line.

Figure 3: Voteshare vs Presvote.



3. Write the prediction equation.

$$\hat{Y} = \alpha_3 + \beta_3 X = 0.441 + 0.388X$$

$$\widehat{\texttt{voteshare}} = 0.441 + 0.388\texttt{presvote}$$

In R:

```
alpha_rl_3 <- reg_line_3$coefficients[1]
beta_rl_3 <- reg_line_3$coefficients[2]

# y = alpha_rl_3 + beta_rl_3*x

p_vs_fun_3 <- function(x) {
  y <- alpha_rl_3 + beta_rl_3*x
  return(y)
}
```

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

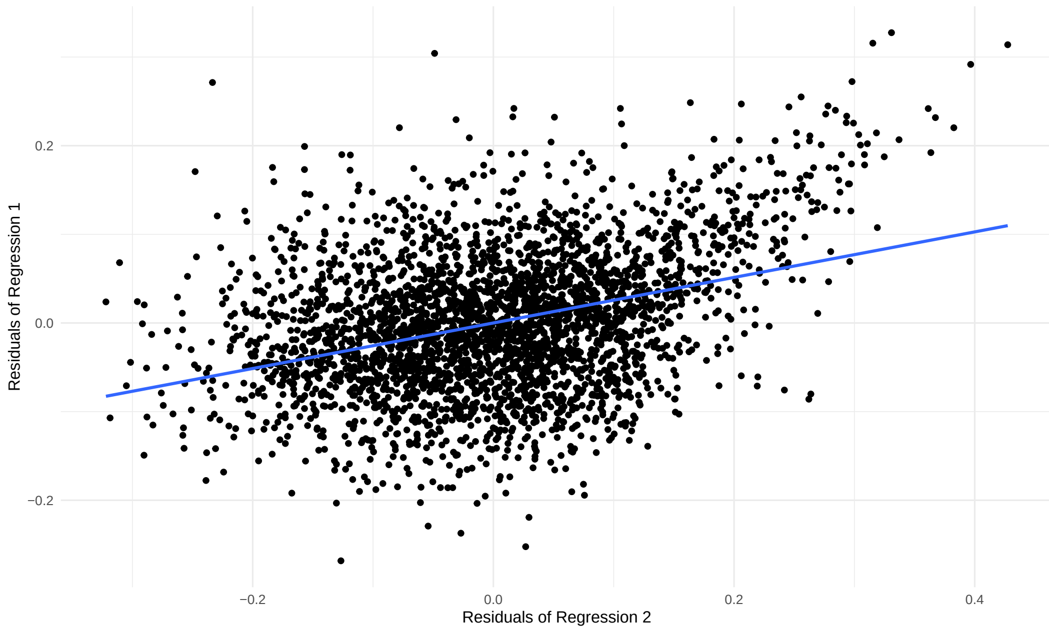
```r
reg_line_4 <- lm(residuals_rl_1 ~ residuals_rl_2)
```

Table 4: Residuals of Reg 1 Explained by Residuals of Reg 2

	Dependent variable:
	Residuals Reg 1 Coefficients (Reg 4)
Residuals Reg 2	0.257***
	(0.012)
Constant	−0.000
	(0.001)
Observations	3,193
R^2	0.130
Adjusted R^2	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Make a scatterplot of the two residuals and add the regression line.

Figure 4: Residuals 1 vs Residuals 2.



3. Write the prediction equation.

$$\hat{Y} = \alpha_4 + \beta_4 X = -1.942 \times 10^{-18} + 0.388X$$

$$\widehat{\texttt{residuals}}\ \texttt{RL 1} = -1.942 \times 10^{-18} + 0.388\texttt{residuals RL 2}$$

In R:

```
alpha_rl_4 <- reg_line_4$coefficients[1]
beta_rl_4 <- reg_line_4$coefficients[2]

# y = alpha_rl_4 + beta_rl_4*x

p_vs_fun_4 <- function(x) {
  y <- alpha_rl_4 + beta_rl_4*x
  return(y)
}
```

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
reg_line_5 <- lm(inc_sub$voteshare ~ inc_sub$difflog + inc_sub$presvote,
    data = inc_sub)
```

Table 5: Incumbent Vote Share Explained by Spending and Presidential Vote Share

	Dependent variable:
	Voteshare Coefficients (Reg 5)
Spending (Difflog)	0.036***
	(0.001)
Presvote	0.257***
	(0.012)
Constant	0.449***
	(0.006)
Observations	3,193
R^2	0.450
Adjusted R^2	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
Note:	*p<0.1; **p<0.05; ***p<0.01

2. Write the prediction equation.

$$\hat{Y} = \alpha_5 + \beta_{difflog}X_{difflog} + \beta_{presvote}X +_{presvote} = 0.449 + 0.036X_{difflog} + 0.257X_{presvote}$$

$$\widehat{\texttt{voteshare}} = 0.449 + 0.036\texttt{difflog} + 0.257\texttt{presvote}$$

In R:

```r
alpha_rl_5 <- reg_line_5$coefficients[1]
beta_rl_5_difflog <- reg_line_5$coefficients[2]
beta_rl_5_presvote <- reg_line_5$coefficients[3]
# predicted voteshare = alpha_rl_5 + beta_rl_5_difflog*x1 + beta_rl_5_presvote*x2

p_vs_fun_5 <- function(diff_log, pres_vote) {
  y <- alpha_rl_5 + beta_rl_5_difflog*diff_log + beta_rl_5_presvote*pres_vote
  return(y)
}
```

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The slope (and standard error) of the Presidential Vote Share variable in the multivariate regression above is identical to the slope (and standard error) of the Residuals 2 variable (residuals of Presidential Vote Share $\sim$ Spending) when used to explain variation in the Residuals 1 variable (residuals of Incumbent Vote Share $\sim$ Spending). This makes sense, as, in steps 2 and 3, we isolate and remove the impact of Spending on Incumbent Vote Share and Presidential Vote Share to illuminate the individual impact of Presidential Vote Share on Incumbent Vote Share (holding Spending constant).